4.2
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1) (a) Dom. = $\{p \in P \mid p$ is parent$\}$
Range = $\{q \in P \mid q$ is a child$\}$ or q has living parent.

(b) $\{(x,y) \in R^2 \mid y > x^2\}$
Dom = $\{x \in R \mid x^2\}$ or Dom = R
~~Range $\{y \in R^+ \mid y\}$~~ Range = R^+ (as x^2 is +ve, so any $y > x^2$ will be positive)
Every x^2 has some y & vice versa

2) (a) Dom = $\{p \in P \mid p$ is someone's brother$\}$
Range $\{p \in P \mid p$ has a brother$\}$
(b) Dom = $\{x \in R^+ \mid x \geq 1\}$ as solving it for y give
$y = \sqrt{1 - \frac{2}{x^2+1}}$ and for this $\sqrt{}$ to be +ve
$\frac{2}{x^2+1}$ must be ≤ 1
~~If its not we will have complex numbers not R.~~
Range = R^+ (square root of +ve number is +ve)

3) $L = \{(s,r) \in S \times R\}$ $L^{-1} = \{(r,s) \in R \times S\}$
$E = \{(s,c) \in S \times C\}$ $E^{-1} = \{(c,s) \in C \times S\}$
$L^{-1} \circ L = \{(s,t) \in S \times S \mid \exists r \in R((s,r) \in R \wedge (r,t) \in L^{-1}\}$
$= \{(s,t) \in S \times S \mid$ student s lives in room r & so is student t$\}$
(b) $E \circ (L^{-1} \circ L)$ $L^{-1}L = S \times S$
$(s,c) \times (s,s) = S \times C$
$E \circ (L^{-1} \circ L) = \{(s,c) \in S \times C \mid \exists s \in S (s,c) \in E \wedge (s,s) \in L^{-1} \circ L\}$
$= \{(s,c) \in S \times C \mid$ some student s is enrolled in course c and lives in room r$\}$
$\{$ some student s who lives in some room is enrolled in course c $\}$

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4) $E = \{(s,c) \in S \times C\}$

$T = \{(c,p) \in C \times P\}$ $T^{-1} = \{(p,c) \in P \times C\}$

$M = \{(c,d) \in C \times D \mid$ The course c meets on day $d\}$ for $D = \{$Mon To Fri$\}$

(a) $M \circ E = \{(s,d) \in S \times D \mid \exists c \in C((s,c) \in E \wedge (c,d) \in M))\}$

$= \{(s,d) \in S \times D \mid \exists c \in C$ (student s is enrolled in some course c that meets on day d)$\}$

= " | student s is enrolled in some course that meets on day $d\}$

(b) $M \circ T^{-1} = \{(p,d) \in P \times D \mid \exists c((p,c) \in T^{-1} \wedge (c,d) \in M)\}$

$= \{(p,d) \in P \times D \mid \exists c \in C(((c,p) \in T \wedge (c,d) \in M)\}$

$= \{(p,d) \in P \times D \mid$ some course c is taught by professor p and meets on day $d\}$

$(B \times M) \circ (A \times B)$ These 2 must be same

5) (a) $S \circ R = \{(a,b) \in A \times B \mid \exists b \in B (a, b) \in R \wedge (b, b) \in S\}$

These b may or may not be same

~~$S \circ R = \{(3,6)$~~

Note that we have to pick (a,b) from $A \times B$.

$A \times B = \{(1,4),(1,5),(2,4),(2,5),(3,4),(3,5)\}$
$(1,6)$ $(2,6)$ $(3,6)$

$S \circ R = \{(1,5),(1,6),(1,4),(2,4),(3,6)\}$

~~$(1,4) \in R$ & $(4,6) \in S$~~ $(1,5) \in R$ & $(5,4) \in S$.

(b) $S \circ S^{-1}$ $S^{-1} = \{$~~$(4,4),(5,4),(6,5$~~
$B \times B$ $B \times B$
$(5,4),(6,4),(4,5),(6,6)\}$

$S \circ S^{-1} = \{(5,5),(5,6),(6,5),(6,6),(4,4),$ ~~$(6,6)$~~ $\}$

explanation of (b) suppose $S = \{(x,y)\} \subseteq B \times B$
& $S^{-1} = (y,x) \in B \times B$
$S \circ S^{-1} = \{(y,y) \in B \times B \mid \exists x((y,x) \in S^{-1} \wedge (x,y) \in S\}$

6) $S^{-1} \circ R$ $(C \times B) \circ (A \times C) = A \times B$

$S^{-1} = \{(7,4), (8,4), (6,5)\}$

$S^{-1} \circ R = \{(1,4), (3,5), (3,4)\}$

(b) $R^{-1} \circ S$ $(C \times A) \circ (B \times C) = B \times A$

$R^{-1} = \{(7,1), (6,3), (7,3)\}$

$R^{-1} \circ S = \{(4,1), (4,3), (5,3)\}$

7) (a) $Ran(R^{-1}) = Dom(R)$

$b \in Ran(R^{-1}) \longleftrightarrow p \in Dom(R)$

$R \subseteq A \times B$, $R^{-1} \subseteq B \times A$ $b \in Dom(R)$

~~$(a,b) \in R$ iff $(b,a) \in R^{-1}$~~

~~$(b,a) \in R^{-1}$ means $\exists$~~

~~supp $a \in Ran(R^{-1})$~~

~~$\exists b\ (a,b) \in R^{-1}$~~

$b \in Ran(R^{-1})$ iff $\exists a \in A\ (a,b) \in R^{-1}$

so $a \in A$ & $(a,b) \in R^{-1}$

so $(b,a) \in R$

so $b \in Dom(R)$ as $\exists a \in A\ (b,a) \in R$.

(b) $(R^{-1})^{-1} = R$ - ① $Ran(R^{-1}) = Dom(R)$

$Dom(R^{-1}) = Ran(R)$ ②

By a fact $Dom(R) = Dom(R^{-1})^{-1}$ by ①

$= Ran(R^{-1})$ by ②

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(c) $T \circ (S \circ R) = (T \circ S) \circ R$

As both are relations fro $A \times D$.

so let $(a,d) \in (T \circ S) \circ R$ — ① $(a,d) \in T \circ (S \circ R)$

② $R \subseteq A \times B$

$S \subseteq B \times C$

$T \subseteq C \times D$

$S \circ R = (B \times C) \circ (A \times B) = A \times C$

$T \circ (S \circ R) = ((C \times D) \circ (A \times C) = A \times D$

~~① means~~ $\exists$

$(T \circ S) = (C \times D) \circ (B \times C) = B \times D$

$(T \circ S) \circ R = (B \times D) \circ (A \times B) = A \times D$.

① mes $\exists b((a,b) \in R \wedge (b,d) \in T \circ S)$

and $(b,d) \in T \circ S$ mes

In sum $\exists b[(a,b) \in R \wedge \exists c((b,c) \in S \wedge (c,d) \in T)]$

Now we can apply existential inst

$(a,b) \in R$, $(b,c) \in S$, $(c,d) \in T$

as $(a,b) \in R$ & $(b,c) \in S$ so $(a,c) \in S \circ R$

And as $(a,c) \in S \circ R$ & $(c,d) \in T$

so $(a,d) \in T \circ (S \circ R)$ our goal.

Our goal was actually

$(a,d) \in T \circ (S \circ R)$

~~$\exists b[(a,b) \in R$~~ $(a,c) \in S \circ R]$

$\exists c[(c,d) \in T \wedge (a,c) \in S \circ R$

$\exists c[(c,d) \in T \wedge \exists b((b,c) \in S \wedge (a,b) \in R)]$

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(d) $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$

$R \subseteq A \times B$ $R^{-1} \subseteq B \times A$

$S \subseteq B \times C$ $S^{-1} \subseteq C \times B$

$\rightarrow$ $S \circ R \subseteq A \times C$

so let $(a,c) \in S \circ R$ so $(c,a) \in (S \circ R)^{-1}$ $(c,a) \in R^{-1} \circ S^{-1}$

$\exists b[(a,b) \in R \wedge (b,c) \in S]$ as $R^{-1} \circ S^{-1} \subseteq C \times A$

by existential inst $\exists b[(c,b) \in S^{-1} \wedge (b,a) \in R^{-1}]$

$(a,b) \in R$ $\longrightarrow$ $\exists b[(b,c) \in S \wedge (a,b) \in R]$

$(b,c) \in S$

actually this was $(b,a) \in R^{-1}$ & $(c,b) \in S^{-1}$ so

so $(c,a) \in R^{-1} \circ S^{-1}$ ~~(a,c)~~

As $(a,c) \in S \circ R$ iff $(c,a) \in (S \circ R)^{-1}$ are equivalent statements, so we can use definition of $(a,c) \in S \circ R$ instead of other.

$\leftarrow$ $(c,a) \in R^{-1} \circ S^{-1}$

$\exists b[(b,c) \in S \wedge (a,b) \in R]$ $(c,a) \in (S \circ R)^{-1}$

$(a,c) \in R \circ S$

$\longrightarrow$ $\exists b[(a,b) \in R \wedge (b,c) \in S]$

$(c,b) \in S^{-1}$

$(b,a) \in R^{-1}$ so ~~(c,a)~~

As $(b,c) \in S \wedge (a,b) \in R$ so $(a,c) \in S \circ R$ so $(c,a) \in (S \circ R)^{-1}$

~~Here's a clear proof by using equivalences.~~

Supp $(c,a) \in (S \circ R)^{-1}$ iff $(a,c) \in S \circ R$

iff $\exists b \in B[(a,b) \in R \wedge (b,c) \in S]$

iff $\exists b \in B[(c,b) \in S^{-1} \wedge (b,a) \in R^{-1}]$

iff $(c,a) \in R^{-1} \circ S^{-1}$

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8) $E = \{(p, q) \in P \times P \mid p$ is envy of $q\}$

$E \circ E$ is $\{(p_1, p_2) \in P \times P \mid \exists q((p_1, q) \in E \wedge (q, p_2) \in E)\}$

we change p, q to p, p₂ because there some elements are being equal — By definition of composition

e.g. E∘E mus ∃q so we have to change the name of variables for better understand

iff (q, p) (p, q)

Now E∘E $\{(p_1, p_2) \in P \times P \mid p_1$ is envy of q & q is envy of $p_2\}$

" $\mid p_1$ is envy of someone who is envy of $p_2\}$

But (p_1, p_2) must be friends, so $(p_1, p_2) \in F$

so $E \circ E \subseteq F$ is our required translation

~~we were able~~

9) $R \subseteq A \times B$

$S \subseteq B \times C$

(a) $Dom(S \circ R) \subseteq Dom(R)$

Note that $S \circ R \subseteq A \times C$

$\forall a [a \in Dom(S \circ R) \longrightarrow a \in Dom(R)]$

Supp $a \in Dom(S \circ R)$ — $a \in Dom R$

iff $\exists c \in C (a, c) \in S \circ R$ — iff $\exists b \in B (a, b) \in R$

$(a, c) \in S \circ R$ iff $\exists b (a, b) \in R$

$\wedge (b, c) \in S$

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$\forall b[b \in Ran(R) \to b \in Dom(S)]$

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(b) $Ran(R) \subseteq Dom(S)$ ① $\longrightarrow$ $Dom(S \circ R) = Dom\,R$

We already prove in part(a) that $Dom(S \circ R) \subseteq Dom(R)$.

$\leftarrow$ Supp $a \in Dom(R)$ $\quad a \in Dom(S \circ R)$

so $\exists b \in B\,(a,b) \in R$ $\quad \exists b \in B[(a,b) \in R \wedge (b,c) \in S]$

$b_1 \in B\,(a,b_1) \in R$:- existential inst

Also we can see that $b_1 \in Ran(R)$

so by ① $b_1 \in Dom(S)$

iff $\exists c \in C\,(b_1,c) \in S$

since $(a,b_1) \in R$ & $(b_1,c) \in S$ so $(a,c) \in S \circ R$ so

$a \in Dom(S \circ R)$.

(c) $Ran(S \circ R) \subseteq Ran(S)$ & if $Dom(S) \subseteq Ran(R)$ then $Ran(S \circ R) = Ran(S)$

Suppose $c \in Ran(S \circ R)$ $\quad c \in Ran(S)$

$\exists a \in A\,(a,c) \in S \circ R$ $\quad \exists b \in B(b,c) \in S$

$\exists b \in B[(a,b) \in R \wedge (b,c) \in S]$

Now we have that $Ran(S \circ R) \subseteq Ran(S)$ so to prove the second theorem, we will prove $\leftarrow$ side only

Supp $Dom(S) \subseteq Ran(R)$ $\quad Ran(S \circ R) = Ran(S)$

$\forall b[b \in Dom(S) \to b \in Ran(R)]$ ① $\forall c[c \in Range(S \circ R) \leftrightarrow c \in Ran(S)]$

$\leftarrow$ Supp $c \in Ran(S)$ $\quad c \in Ran(S \circ R)$

then $\exists b \in B\,(b,c) \in S$ $\quad$ ~~$\exists b \in B[(a,b) \in R \wedge (b$~~

$b_1 \in B\,(b_1,c) \in S$ $\quad$ iff $\exists a \in A\,(a,c) \in S \circ R$

so $b_1 \in$ ~~Ran~~ Dom (S) $\quad$ iff $\exists b \in B[(a,b) \in R \wedge (b,c) \in S]$

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so by ① $b_1 \in Ran(R)$

$\exists a \in A\ (a, b_1) \in R$

so we find $a_1 \in A\ [\, b_1 \in B\ (a, b_1) \in R \land (b_1, c) \in S \,]$

10) $R \subseteq A \times B$

$S \subseteq A \times B$

(a) $R \subseteq Dom(R) \times Ran(R)$

supp $(a, b) \in R$ — $(a, b) \in Dom(R) \times Ran(R)$

$a \in Dom(R),\ b \in Ran(R)$ — $a \in Dom\,R \land b \in Ran(R)$

(b) $R \subseteq S \longrightarrow R^{-1} \subseteq S^{-1}$

$\forall (a, b) \in A \times B\ [(a, b) \in R \to (a, b) \in S]$ ① — $\forall (b, a) \in B \times A\ [(b, a) \in R^{-1} \to (b, a) \in S^{-1}]$

$(b, a) \in S^{-1}$

supp $(b, a) \in R^{-1}$

iff $(a, b) \in R$

by ① $(a, b) \in S$ iff $(b, a) \in S^{-1}$

(c) $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$

All 3 of these statement are True

supp $(b, a) \in (R \cup S)^{-1}$

iff $(a, b) \in R \cup S$

iff $(a, b) \in R \lor (a, b) \in S$

iff $(b, a) \in R^{-1} \lor (b, a) \in S^{-1}$

iff $(b, a) \in R^{-1} \cup S^{-1}$ Proved.

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11) $R \subseteq A \times B$

$S \subseteq B \times C$

$S \circ R = \phi$ iff $Ran(R) \cap Dom(S) = \phi$

$\rightarrow$ $\sim \exists (a,c) \in A \times C \, [(a,c) \in S \circ R]$ $\forall b [b \in Ran(R) \rightarrow b \notin Dom(S)]$

$\forall (a,c) \in A \times C \sim [(a,c) \in S \circ R]$ Let b be arbitrary

$\forall (a,c) \sim [\exists b \in B ((a,b) \in R \wedge (b,c) \in S)]$

$\forall (a,c) \in A \times C [\forall b \in B ((a,b) \notin R \vee (b,c) \notin S]$ ①

Supp $b \in Ran(R)$ $b \notin Dom(S)$

that $\exists a \in A \; (a,b) \in R$

Since $(a,b) \in R$ for arbitrary b so by ① $(b,c) \notin S$ so

$b \notin Dom(S)$.

Proof by contradiction was easier here since goal is basically a negative statement.

$\leftarrow$

$Ran(R) \cap Dom(S) = \phi$ $S \circ R = \phi$

$\forall b [b \in Ran(R) \rightarrow b \notin Dom(S)]$

Supp $S \circ R \neq \phi$ Contradict

then $\exists (a,c) \in A \times C \, [(a,c) \in S \circ R]$

iff $\exists b \; (a,b) \in R \wedge (b,c) \in S$

so lets say $b = b_1$ then $(a,b_1) \in R$

$\wedge (b_1,c) \in S$

so $b_1 \in Ran(R) \wedge b_1 \in Dom(S)$ which is a contradiction

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12) $R \subseteq A \times B$

$S \subseteq B \times C$, $T \subseteq B \times C$

$(S \circ R) \setminus (T \circ R) \subseteq (S \setminus T) \circ R$

$\forall (a,c)[(a,c) \in (S \circ R) \setminus (T \circ R) \longrightarrow (a,c) \in (S \setminus T) \circ R]$

Supp $(a,c) \in S \circ R \wedge (a,c) \notin T \circ R$ $\qquad (a,c) \in (S \setminus T) \circ R$

$(a,c) \in S \circ R$ iff $\exists b \in B[(a,b) \in R \wedge (b,c) \in S]$ ①

$(a,c) \notin T \circ R$ iff $\sim \exists b \in B[(a,b) \in R \wedge (b,c) \in T]$

$\forall b \in B[(a,b) \notin R \vee (b,c) \notin T]$ ②

by ① $(a,b_1) \in R \wedge (b_1,c) \in S$

by ② as $(a,b_1) \in R$ so $(b_1,c) \notin T$

(② can be rewritten in form of $p \to q$)

$(a,c) \in$ iff $\exists b \in B[(a,b) \in R \wedge (b,c) \in S \setminus T]$

$\exists b \in B[(a,b) \in R \wedge (b,c) \in S \wedge (b,c) \notin T]$

goal follows

(b) The mistake is that we have existential statement $(a,b_1) \in R$, $(b_1,c) \in S$ and $(b_1,c) \notin T$. We cannot conclude a universal statement from this, that is $\forall b \in B[(a,b) \notin R \vee (b,c) \notin T]$ there can be another b_2 that makes it false

(c) contradiction let $A: \{1,2\}$, $B: \{2,3\}$, $C: \{3,4\}$

& $R: \{(1,2),(1,3)\}$, $S: \{(2,3)\}$, $T: \{(3,3)\}$

then $(S \setminus T) \circ R: \{(1,3)\}$

& $(S \circ R) \setminus (T \circ R) = \{(1,3),(1,4)\}$.

Note we found contradiction by choosing $b_2 \in B$ such that $(a,b_2) \in R \wedge (b_2,c) \in T$

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13) $R \subseteq A \times B$

$S \subseteq A \times B$

$T \subseteq B \times C$

(a) $R \cap S = \phi \longrightarrow R^{-1} \cap S^{-1} = \phi$

$\forall (a,b) \in A \times B \, [(a,b) \in R \rightarrow (a,b) \notin S]$

~~Contradict~~

supp $R^{-1} \cap S^{-1} \neq \phi$

$R \cap S \neq \phi$

then $\exists (b,a) \in R^{-1}$ & S^{-1}

so $(a,b) \in R \wedge (a,b) \in S$ so $R \cap S \neq \phi$

So this is proof by contraposition

(b) $R \cap S = \phi$ $(T \circ R) \cap (T \circ S) = \phi$

$\forall (a,b) \, [(a,b) \in R \rightarrow (a,b) \notin S]$

Proof by Contraposition:

supp $(T \circ R) \cap (T \circ S) \neq \phi$ $R \cap S \neq \phi$

$\exists (a,c) \, [(a,c) \in T \circ R \wedge (a,c) \in T \circ S]$

$[(a_1,c_1) \in T \circ R \wedge (a_1,c_1) \in T \circ S]$ — These are 2 separate give

~~$\exists b [(a_1,b) \in R \wedge (b,c_1) \in T \wedge (a_1,b) \in S \wedge (b,c_1) \in S]$~~

~~so now as $(a_1,b_1) \in R \wedge (a_1,b_1) \in S$ so $R \cap S \neq \phi$~~

$\exists b ((a_1,b) \in R \wedge (b,c_1) \in T]$ $\exists b ((a_1,b) \in S \wedge (b,c_1) \in T]$ - These 2 bs may or may not be same

The key here is that existential quantifier ~~doesnt distribute~~ ~~over conjunction~~

— This is also False ~~theorem~~ statement.

(c) $T \circ R \cap T \circ S = \phi$ $R \cap S = \phi$

$\forall (a,c) \, [(a,c) \in T \circ R \rightarrow (a,c) \notin T \circ S]$

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Counterexample to (b) ~~A = ... , B =~~ $A = \{1,2\}$, $B = \{2,3\}$
$C = \{3,4\}$
$R = \{(1,2)\}$, $S = \{(1,3)\}$, $T = \{(2,3), (3,3)\}$
NOW $T \circ R = \{(1,3)\}$ & $T \circ S = \{(1,3)\}$

Counterexample to (c). $R = \{(1,2)\}$, $S = \{(1,2)\}$, $T = \{(3,3)\}$
Now $T \circ R$ & $T \circ S = \phi$ so they are disjoint but R&S are not

14) $R \subseteq A \times B$
$S \subseteq B \times C$, $T \subseteq B \times C$

(a) $S \subseteq T \longrightarrow S \circ R \subseteq T \circ R$

$\forall (b,c) \in B \times C\, [(b,c) \in S \rightarrow (b,c) \in T]$ 1 $\forall (a,c)\, [(a,c) \in S \circ R \rightarrow (a,c) \in T \circ R]$

Supp $(a,c) \in S \circ R$ $(a,c) \in T \circ R$

so $\exists b \in B\, [(a,b) \in R \wedge (b,c) \in S]$ $\exists b \in B\, [(a,b) \in R \wedge (b,c) \in T]$

we have $b_1 \in B$ $(a,b_1) \in R$: contradiction

& so $(b_1, c) \in S$ so by 1 $(b_1, c) \in T$ hence proved

(b) $(S \cap T) \circ R \subseteq (S \circ R) \cap (T \circ R)$

Supp $(a,c) \in (S \cap T) \circ R$ $(a,c) \in (S \circ R) \cap (T \circ R)$

$\exists b \in B\, [(a,b) \in R \wedge (b,c) \in S \wedge (b,c) \in T]$ 1

$b_1 \in B$ $(a,c) \in S \circ R \wedge (a,c) \in T \circ R$

$(a,b_1) \in R$ $\exists b \in B\, [(a,b) \in R \wedge (b,c) \in S]$ 2

$(b_1,c) \in S$ $\exists b \in B\, [(a,b) \in R \wedge (b,c) \in T]$ 3

$(b_1,c) \in T$

Proved.

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(c) $(S \cap T) \circ R = (S \circ R) \cap (T \circ R)$

We already know $\rightarrow$ side by part (a)

$\leftarrow$ Supp

$(a, b_1) \in R \wedge (b_1, c) \in S$ :: existential inst for ②

$(a, b_2) \in R \wedge (b_2, c) \in T$ :: existential inst for ③

Goal does not follow as b_1 & b_2 can be different.

Counterexample:

$R = \{(1,3), (1,4)\}$ $S = \{(3,5)\}$, $T = \{(4,5)\}$.

(d) $(S \cup T) \circ R = (S \circ R) \cup (T \circ R)$

$\rightarrow$ Supp $(a, c) \in (S \cup T) \circ R$

$\exists b \in B[(a,b) \in R \wedge ((b,c) \in S \vee (b,c) \in T)]$

$b_1 \in B$

$(a, b) \in R$

$(b_1, c) \in S \vee (b_1, c) \in T$

In both cases, goal follows clearly

$(a, c) \in S \circ R \vee (a,c) \in T \circ R$

$\exists b \in B[(a,b) \in R \wedge (b,c) \in S] \vee$

$\exists b \in B[(a,b) \in R \wedge (b,c) \in T]$

$\leftarrow$ Supp $\exists b \in B[(a,b) \in R \wedge (b,c) \in S]$ ①

$\vee \exists b \in B[(a,b) \in R \wedge (b,c) \in T]$

we have to prove that

$\exists b \in B[(a,b) \in R \wedge ((b,c) \in S \vee (b,c) \in T)]$

$b_1 \in B, (a, b_1) \in R$

& $(b_1, c) \in S$ ①

$b_2 \in B$ $(a, b_2) \in R$

$(b_2, c) \in T$ ②

Goal follows in both cases

Mergestack

Date: ___ / ___ / ___ MON — TUE — WED — THUR — FRI — SAT — SUN

15) $R \subseteq A \times B$ — we can prove this easily

$S \subseteq C \times D$

$\exists E \, [(R \subseteq A \times E) \wedge (S \subseteq E \times D)]$

Any set E with $B \cup C \subseteq E$ works.

Clearly $R \subseteq A \times (B \cup C)$ as $R \subseteq A \times B$ — Any $x \in B \cap C$ will satisfy this & so this cut can be applied

and $S \subseteq (B \cup C) \times D$ as $S \subseteq C \times D$

& so $R \subseteq A \times E$ $\exists x \, ((a, x) \in A \times (B \cup C) \wedge (x, d) \in (B \cup C) \times D)$

For any $(a, b) \in R \subseteq A \times B$. For $R \subseteq A \times E$, any b in (a, b) must be in E too, so $B \subseteq E$. Similarly $C \subseteq E$ so $B \cup C \subseteq E$.

The actual xs that will make the bridge between R & S will be from $B \cap C$. But minimum requirement of E is $B \cup C \subseteq E$.

~~Because~~ as $R \subseteq A \times B$ & if $(a, x) \in A \times B \cup C$ then $x \in B \cup C$ & we know $R \subseteq A \times B$ so $x \in B$. For S, $x \in C$, so $x \in B \cap C$.

~~But to formally write $R \subseteq A \times E$ we need $B \cup C$. Because for $B \cap C$~~ ~~R may not remain $\subseteq A \times E$ as $B \cap C$ is not guaranteed~~ to include all B.

$\forall (x, y) \, [(a, b) \in R \rightarrow (a, b) \in B \cup C]$ — prove $a \in A$ & $b \in B \cup C$

supp $(a, b) \in R$

so $a \in A$ & $b \in B$

clearly $a \in A$ & $b \in B$ so $b \in B \cup C$